

CineScribe

High-Fidelity Style-Conditioned Video Captioning Agent

AMD Developer Hackathon 2026

Track 2 Submission | public registry: [urlsandcodes/cinescribe](https://urlsandcodes.com/cinescribe)

GitHub: github.com/urlsandcodes/cinescribe

Project Goal & Challenge

The Core Challenge: Automatically describing video clips across diverse styles (Formal, Sarcastic, Humorous Tech, and Humorous Non-Tech) without compromising factual accuracy.

- **Factual Grounding:** Captions must accurately reflect specific details (subjects, actions, background objects, weather, OCR text) without hallucinating.
- **Stylistic Tone Fidelity:** Tone variants must sound naturally distinct (sarcastic should be sharp/ironic; humorous tech must weave in computing jargon; humorous non-tech must be clean, everyday humor).
- **Temporal Fusion:** Correctly stitching visual sequences and audio transcripts chronologically to formulate complete context.

2-Stage Visual-Speech Fusion Architecture

CineScribe splits representation from generation to enforce 100% accuracy and style alignment:

Stage 1: Factual Extraction	Stage 2: Stylistic Translation
• Segment video using scene change threshold. • Extract representative midpoint frame for each scene. • Run VLM descriptor on frames (Subject, Actions, Background, OCR). • Run Whisper to transcribe spoken dialog (if audio exists).	• Ingest chronological visual + audio timeline event sequence. • Prompt DeepSeek V4 Flash with tone context and few-shot examples. • Run generation at 0.6 temperature for creative style fitting. • Parse resulting JSON captions matching target schema.

Technical Choices & Rationale

Why Fireworks Serverless Models? Bypass container deployment complexity. Zero cold-starts, high scalability, and robust response times under 15 seconds.

Component	Selected Choice	Rationale / Advantages
VLM (Vision)	Kimi K2.6	A powerful reasoning-based vision model. Outputting its chain-of-thought ensures high-precision details of subject, actions, and background, which prevents VLM hallucination.
LLM (Captioning)	DeepSeek V4 Flash	Ultra-fast, cost-effective model that excels at complex instruction following. Natural tone synthesis with strict JSON schema compliance.
Audio	Whisper API	High-accuracy voice transcription to seamlessly capture speech cues alongside visual actions.

Technical Specs & Submission Compliance

CineScribe has been engineered to perfectly comply with all hackathon agent specifications:

- **Harness Architecture:** Built as a **linux/amd64** Docker image, resolving Apple Silicon QEMU segfaults by using standard pip modules.
- **Input/Output Mounts:** Reads video tasks directly from ``/input/tasks.json`` and writes valid output to ``/output/results.json`` before exiting.
- **Performance Constraints:** Starts up in **< 60 seconds**, processes clips dynamically in under 30 seconds per request, and has an exit code of 0 on success.
- **Robust Failover:** Handles local rate limits with exponential retries and automatically cleans up local clips after execution.